

# IPTRC OPS COPILOT

ServiceNow Deployment Guide

Version 1.0 | AT&T IPTRC | 2026-05-28

|                 |                                                   |
|-----------------|---------------------------------------------------|
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| Audience:       | IPTRC Tooling Engineers + Operations Stakeholders |
| Document Type:  | Technical Deployment Guide                        |
| Classification: | AT&T; Internal — Do Not Distribute Externally     |
| Status:         | DRAFT — Review before deployment                  |

■ This document contains operational guidance and code. Never hardcode API keys. Always store credentials in ServiceNow Credential Store. All AI outputs are DRAFTS — engineer review required before operational use.

## Executive Summary

IPTRC Ops Copilot is an AI-powered assistant built for AT&T; IPTRC network engineers. It reduces operational toil by converting unstructured inputs — CLI outputs, email text, ticket descriptions — into structured, copy/paste-ready artifacts. This guide covers the complete ServiceNow deployment including Script Include, Scripted REST API, Service Portal Widget, and system configuration.

## Skill Modules

| SKILL A Ticket Notes Generator |                                                                            |
|--------------------------------|----------------------------------------------------------------------------|
| What it does:                  | Produces structured, audit-safe work notes from CLI dumps and ticket text. |
| Inputs:                        | Platform, Ticket ID, Severity, Symptoms, CLI Output, Actions Taken         |
| Output:                        | Work Notes with Summary / Observations / Actions / Next Steps / Risks      |

**SKILL B    SMOP / GMOP Scaffolder**

|                      |                                                                                               |
|----------------------|-----------------------------------------------------------------------------------------------|
| <b>What it does:</b> | Drafts a SMOP/GMOP skeleton with pre/post checks, command bundles, and rollback placeholders. |
| <b>Inputs:</b>       | Hostname, Platform, OS Version, Work Type, Maintenance Window, Timezone                       |
| <b>Output:</b>       | DRAFT scaffold with all required MOP sections — always labeled DRAFT                          |

**SKILL C    RMA Email Composer**

|                      |                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------|
| <b>What it does:</b> | Generates a KGPCo-compliant text-only RMA email with all required fields in correct order. |
| <b>Inputs:</b>       | AOTS Ticket, Platform, Problem Description, PID/SN, CLLI, OSWF, Shipping, CLI snapshots    |
| <b>Output:</b>       | TEXT-ONLY email body ready to send — subject line included                                 |

**SKILL D    Maintenance Email Summarizer**

|                      |                                                                                                   |
|----------------------|---------------------------------------------------------------------------------------------------|
| <b>What it does:</b> | Extracts window, timezone, circuits, and produces an IPTRC action checklist from carrier notices. |
| <b>Inputs:</b>       | Paste full carrier maintenance notification email body                                            |
| <b>Output:</b>       | Window / Timezone / Affected Circuits + IPTRC Actions Checklist                                   |

## Architecture Overview

| Layer  | Component                       | Description                                                                      |
|--------|---------------------------------|----------------------------------------------------------------------------------|
| UI     | Service Portal Widget           | AngularJS tabbed interface — 4 skill tabs, form inputs, copy-to-clipboard output |
| API    | Scripted REST API               | 4 POST endpoints, one per skill — no auth bypass, uses session context           |
| Logic  | IPTRCOpsSkills (Script Include) | Core AI engine — sanitization, LLM call, output validation, auto-repair          |
| AI     | Azure OpenAI / Anthropic        | External LLM via sn_ws.RESTMessageV2 — credentials in Credential Store           |
| Config | System Properties               | 4 properties: endpoint, API key, model, max_tokens                               |

■ Call flow: Service Portal Widget → \$http.post → Scripted REST Resource → IPTRCOpsSkills (Script Include) → sn\_ws.RESTMessageV2 → Azure OpenAI → response back through the stack → displayed in widget output area.

## Security Guardrails

- **G-1 No credentials in inputs:** The `_sanitize()` method scrubs 12 regex patterns before any text reaches the LLM — passwords, enable secrets, TACACS keys, RADIUS keys, SNMP auth/priv strings, crypto isakmp keys, API keys, Bearer tokens.
- **G-2 API key in Credential Store:** The `iptrc.ai.apikey` system property should reference a ServiceNow Credential record, never a plaintext key.
- **G-3 Output disclaimers enforced:** SMOP/GMOP output is checked for 'official MOP' language and auto-repaired if found. 'DRAFT' watermark is programmatically prepended if missing.
- **G-4 HTML injection guard:** RMA email output is scanned for HTML tags and repaired to plain text if any are detected.
- **G-5 Heading validation + auto-repair:** All skills validate required output headings. If any are missing, a second LLM call with a repair prompt is made automatically.
- **G-6 No autonomous execution:** This tool generates text artifacts only — it never pushes configs to routers, modifies tickets, or auto-sends emails.

## Deployment Steps

### Step 1: Create System Properties

sys\_properties.list → Create 4 records

|                            |                                                                                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------------|
| <b>iptrc.ai.endpoint</b>   | Your Azure OpenAI endpoint URL (e.g. https://YOUR-RESOURCE.openai.azure.com/openai/deployments/YOUR-DEPLOYMENT) |
| <b>iptrc.ai.apikey</b>     | API key — reference Credential Store record, never plain text                                                   |
| <b>iptrc.ai.model</b>      | Deployment name (e.g. gpt-4o)                                                                                   |
| <b>iptrc.ai.max_tokens</b> | 1500 (adjust per your needs)                                                                                    |

## Step 2: Create Script Include: IPTRCOpsSkills

System Definition → Script Includes → New Name: IPTRCOpsSkills | Accessible from: All application scopes

## Step 3: Create Scripted REST API

Scripted REST APIs → New Name: IPTRC Ops Skills API | API ID: iptrc\_ops\_api Add 4 Resources (POST): /ticket-notes, /smop-draft, /rma-email, /maint-email

## Step 4: Create Service Portal Widget

Service Portal → Widgets → New Name: IPTRC Ops Copilot | Widget ID: iptrc-ops-copilot Paste HTML Template, Client Controller, Server Script, CSS

## Step 5: Add Widget to Portal Page

Service Portal → Pages → [your ops portal page] Drag widget to a container. Set page URL to /sp?id=iptrc\_ops\_copilot

## Step 6: Set REST API Access Controls

Navigate to: System Web Services → Scripted REST APIs → [your API] → Access tab Ensure the API is protected by user authentication (not public)

## Step 7: Test Each Skill

Open the portal widget and test each tab: • Ticket Notes: paste a sample show interface output • SMOP: fill device hostname + platform • RMA Email: fill all required fields • Maint Summary: paste a carrier maintenance email







```
                this._sanitize(p.emailBody||''));
var req = ['MAINTENANCE SUMMARY:', 'Timezone (as stated in email):',
          'IPTRC ACTIONS CHECKLIST:'];
for (var h=0; h<req.length; h++) {
  if (r.text.indexOf(req[h]) === -1 && r.success) {
    r = this._callLLM(sys,
      'MAINTENANCE EMAIL:\n\n' + this._sanitize(p.emailBody||'') +
      '\nMissing headings – rewrite. Timezone must be quoted exactly.');
```

break;

```
    }
  }
  return r;
},

type: 'IPTRCOpsSkills'
};
```



## Component 2 — Scripted REST API Resources

■ Path: System Web Services → Scripted REST APIs → New Name: IPTRC Ops Skills API | API ID: iptrc\_ops\_api | Add 4 POST Resources

### Resource /ticket-notes (POST)

```
(function process(request, response) {
  var body    = request.body.data;
  var skills = new IPTRCOpsSkills();
  var result = skills.generateTicketNotes({
    ticketId:    body.ticketId    || '',
    platform:    body.platform    || '',
    severity:    body.severity    || '',
    symptoms:    body.symptoms    || '',
    cliDump:     body.cliDump     || '',
    actionsTaken: body.actionsTaken || ''
  });
  response.setContentType('application/json');
  response.setStatus(result.success ? 200 : 500);
  response.setBody({ output: result.text, skill: 'ticket_notes' });
})(request, response);
```

### Resource /smop-draft (POST)

```
(function process(request, response) {
  var body    = request.body.data;
  var skills = new IPTRCOpsSkills();
  var result = skills.generateSmopDraft({
    hostname:    body.hostname    || '',
    platform:    body.platform    || '',
    osVersion:   body.osVersion   || '',
    workType:    body.workType    || '',
    windowStart: body.windowStart || '',
    windowEnd:   body.windowEnd   || '',
    timezone:    body.timezone    || '',
    riskNotes:   body.riskNotes   || ''
  });
  response.setContentType('application/json');
  response.setStatus(result.success ? 200 : 500);
  response.setBody({ output: result.text, skill: 'smop_draft' });
})(request, response);
```

### Resource /rma-email (POST)

```
(function process(request, response) {
  var body    = request.body.data;
  var skills = new IPTRCOpsSkills();
  var result = skills.generateRmaEmail(body);
  response.setContentType('application/json');
  response.setStatus(result.success ? 200 : 500);
  response.setBody({ output: result.text, skill: 'rma_email' });
})(request, response);
```

### Resource /maint-email (POST)

```
(function process(request, response) {
  var body    = request.body.data;
  var skills = new IPTRCOpsSkills();
```

```
var result = skills.summarizeMaintenanceEmail({
  emailBody: body.emailBody || ''
});
response.setContentType('application/json');
response.setStatus(result.success ? 200 : 500);
response.setBody({ output: result.text, skill: 'maint_email' });
})(request, response);
```

## Component 3 — Service Portal Widget

■ Path: Service Portal → Widgets → New Name: IPTRC Ops Copilot | Widget ID: iptrc-ops-copilot

### Widget — HTML Template

```
<div class="iptrc-copilot">
  <div class="iptrc-header">
    <h2>IPTRC Ops Copilot</h2>
    <span class="label label-default">AT&T Internal</span>
    <p class="text-muted">AI-generated drafts — review before use.</p>
  </div>

  <!-- Tab navigation -->
  <ul class="nav nav-tabs">
    <li ng-class="{active: c.tab==='ticket'}" ng-click="c.tab='ticket'">
      <a>Ticket Notes</a></li>
    <li ng-class="{active: c.tab==='smop'}" ng-click="c.tab='smop'">
      <a>SMOP / GMOP</a></li>
    <li ng-class="{active: c.tab==='rma'}" ng-click="c.tab='rma'">
      <a>RMA Email</a></li>
    <li ng-class="{active: c.tab==='maint'}" ng-click="c.tab='maint'">
      <a>Maint Summary</a></li>
    </ul>

  <div class="tab-content iptrc-panel">

    <!-- TICKET NOTES TAB -->
    <div ng-show="c.tab==='ticket'">
      <div class="row">
        <div class="col-sm-4">
          <label>Ticket ID (optional)</label>
          <input class="form-control" ng-model="c.ticket.ticketId"
            placeholder="AOTS123456789"/>
        </div>
        <div class="col-sm-4">
          <label>Platform *</label>
          <select class="form-control" ng-model="c.ticket.platform">
            <option value="">Select platform</option>
            <option ng-repeat="p in c.platforms" value="{{p}}">{{p}}</option>
          </select>
        </div>
        <div class="col-sm-4">
          <label>Severity</label>
          <select class="form-control" ng-model="c.ticket.severity">
            <option value="">Select</option>
            <option ng-repeat="s in c.severities" value="{{s}}">{{s}}</option>
          </select>
        </div>
      </div>
      <div class="form-group">
        <label>Symptoms / Ticket Description *</label>
        <textarea class="form-control" rows="4" ng-model="c.ticket.symptoms"
          placeholder="Describe the issue, alarms, customer impact..."></textarea>
      </div>
      <div class="form-group">
        <label>CLI Output / Logs</label>
        <textarea class="form-control iptrc-mono" rows="6"
          ng-model="c.ticket.cliDump"
          placeholder="show interface, show optics, show log..."></textarea>
      </div>
    </div>
  </div>
</div>
```

```

</div>
<div class="form-group">
  <label>Actions Already Taken</label>
  <textarea class="form-control" rows="3"
    ng-model="c.ticket.actionsTaken"
    placeholder="e.g. Rebooted LC, cleared counters..."></textarea>
</div>
<button class="btn btn-primary"
  ng-click="c.runSkill('ticket-notes', c.ticket)"
  ng-disabled="!c.ticket.platform||!c.ticket.symptoms||c.loading">
  <span ng-if="c.loading&&c.activeSkill==='ticket-notes'"
    class="glyphicon glyphicon-refresh spin"></span>
  Generate Work Notes
</button>
</div>

<!-- SMOP/GMOP TAB -->
<div ng-show="c.tab==='smop'">
  <div class="row">
    <div class="col-sm-6">
      <label>Device Hostname *</label>
      <input class="form-control iptrc-mono" ng-model="c.smop.hostname"
        placeholder="e.g. dal13-cr5.ip.att.net"/>
    </div>
    <div class="col-sm-6">
      <label>Platform *</label>
      <select class="form-control" ng-model="c.smop.platform">
        <option value="">Select</option>
        <option ng-repeat="p in c.platforms" value="{{p}}">{{p}}</option>
      </select>
    </div>
    <div class="col-sm-6">
      <label>OS Version</label>
      <input class="form-control" ng-model="c.smop.osVersion"
        placeholder="e.g. IOS-XR 7.9.2"/>
    </div>
    <div class="col-sm-6">
      <label>Work Type *</label>
      <select class="form-control" ng-model="c.smop.workType">
        <option value="">Select</option>
        <option ng-repeat="w in c.workTypes" value="{{w}}">{{w}}</option>
      </select>
    </div>
    <div class="col-sm-4">
      <label>Window Start *</label>
      <input class="form-control" type="datetime-local"
        ng-model="c.smop.windowStart"/>
    </div>
    <div class="col-sm-4">
      <label>Window End *</label>
      <input class="form-control" type="datetime-local"
        ng-model="c.smop.windowEnd"/>
    </div>
    <div class="col-sm-4">
      <label>Timezone *</label>
      <input class="form-control" ng-model="c.smop.timezone"
        placeholder="e.g. CST, UTC"/>
    </div>
  </div>
  <div class="form-group">
    <label>Risk Notes</label>
    <textarea class="form-control" rows="2" ng-model="c.smop.riskNotes"
      placeholder="SLA concerns, customer commitments..."></textarea>
  </div>

```

```

</div>
<button class="btn btn-primary"
  ng-click="c.runSkill('smop-draft', c.smop)"
  ng-disabled="!c.smop.hostname||!c.smop.platform||!c.smop.workType||c.loading">
  <span ng-if="c.loading&&c.activeSkill==='smop-draft'"
    class="glyphicon glyphicon-refresh spin"></span>
  Generate SMOP Scaffold
</button>
<span class="label label-warning">Always outputs as DRAFT</span>
</div>

<!-- RMA EMAIL TAB -->
<div ng-show="c.tab==='rma'">
  <div class="row">
    <div class="col-sm-6">
      <label>AOTS Ticket # </label>
      <input class="form-control iptrc-mono" ng-model="c.rma.aotsTicket"
        placeholder="AOTS123456789012"/>
    </div>
    <div class="col-sm-6">
      <label>Platform </label>
      <select class="form-control" ng-model="c.rma.platform">
        <option value="">Select</option>
        <option ng-repeat="p in c.rmaplatforms" value="{{p}}">{{p}}</option>
      </select>
    </div>
  </div>
  <div class="form-group">
    <label>Problem Description </label>
    <textarea class="form-control" rows="3"
      ng-model="c.rma.problemDescription"
      placeholder="Hardware failure, symptoms, impact..."></textarea>
  </div>
  <div class="form-group">
    <label>Additional Info</label>
    <textarea class="form-control" rows="2"
      ng-model="c.rma.additionalInfo"></textarea>
  </div>
  <div class="row">
    <div class="col-sm-6">
      <label>Defective Part Failure / Requested Part </label>
      <input class="form-control" ng-model="c.rma.defectivePartFailure"/>
    </div>
    <div class="col-sm-6">
      <label>Defective PID / SN </label>
      <input class="form-control iptrc-mono" ng-model="c.rma.defectivePidSn"/>
    </div>
    <div class="col-sm-6">
      <label>Hostname </label>
      <input class="form-control iptrc-mono" ng-model="c.rma.hostname"/>
    </div>
    <div class="col-sm-6">
      <label>FIC Number</label>
      <input class="form-control" ng-model="c.rma.ficNumber"/>
    </div>
    <div class="col-sm-6">
      <label>CLLI </label>
      <input class="form-control iptrc-mono" ng-model="c.rma.clli"
        placeholder="DLSTX123"/>
    </div>
    <div class="col-sm-6">
      <label>OSWF / WMS Ticket </label>
      <input class="form-control iptrc-mono" ng-model="c.rma.oswfTicket"/>
    </div>
  </div>

```

```

    </div>
  </div>
  <div class="form-group">
    <label>Shipping Address *</label>
    <textarea class="form-control" rows="2"
      ng-model="c.rma.shippingAddress"></textarea>
  </div>
  <div class="form-group">
    <label>Onsite Contact (name, number, email) *</label>
    <input class="form-control" ng-model="c.rma.onsiteContact"/>
  </div>
  <div class="row">
    <div class="col-sm-4">
      <label>Vendor Case</label>
      <input class="form-control iptrc-mono" ng-model="c.rma.vendorCase"/>
    </div>
    <div class="col-sm-4">
      <label>NOC Ticket</label>
      <input class="form-control iptrc-mono" ng-model="c.rma.nocTicket"/>
    </div>
    <div class="col-sm-4">
      <label>Same/Next Day *</label>
      <select class="form-control" ng-model="c.rma.sameOrNextDay">
        <option value="">Select</option>
        <option value="Same Day">Same Day</option>
        <option value="Next Day">Next Day</option>
      </select>
    </div>
    <div class="col-sm-4">
      <label>EFA Required? *</label>
      <select class="form-control" ng-model="c.rma.efaRequired">
        <option value="">Select</option>
        <option value="Yes">Yes</option>
        <option value="No">No</option>
      </select>
    </div>
  </div>
  <div class="iptrc-cli-box">
    <h5>CLI Snapshots (paste output)</h5>
    <div class="row">
      <div class="col-sm-6" ng-repeat="f in c.cliFields">
        <label>{{f.label}}</label>
        <textarea class="form-control iptrc-mono" rows="2"
          ng-model="c.rma[f.key]"
          placeholder="Paste {{f.label}} output..."></textarea>
      </div>
    </div>
  </div>
  <button class="btn btn-primary"
    ng-click="c.runSkill('rma-email', c.rma)"
    ng-disabled="c.rmaInvalid()||c.loading">
    <span ng-if="c.loading&&c.activeSkill==='rma-email'"
      class="glyphicon glyphicon-refresh spin"></span>
    Compose RMA Email
  </button>
  <span class="label label-info">Text-only, KGPCo compliant</span>
</div>

<!-- MAINTENANCE EMAIL TAB -->
<div ng-show="c.tab==='maint'">
  <div class="form-group">
    <label>Paste Maintenance Email Body *</label>
    <textarea class="form-control iptrc-mono" rows="14"

```

```

        ng-model="c.maint.emailBody"
        placeholder="Paste full carrier maintenance notification..."></textarea>
    </div>
    <button class="btn btn-primary"
        ng-click="c.runSkill('maint-email', c.maint)"
        ng-disabled="!c.maint.emailBody||c.loading">
        <span ng-if="c.loading&&c.activeSkill==='maint-email'"
            class="glyphicon glyphicon-refresh spin"></span>
        Summarize & Extract Actions
    </button>
</div>

<!-- SHARED OUTPUT AREA -->
<div ng-if="c.output" class="iptrc-output">
    <div class="iptrc-output-header">
        <span class="iptrc-skill-label">{{c.skillLabel}}</span>
        <button class="btn btn-xs btn-default" ng-click="c.copyOutput()">
            <span class="glyphicon"
                ng-class="c.copied ? 'glyphicon-ok' : 'glyphicon-copy'"></span>
            {{c.copied ? 'Copied!' : 'Copy'}}
        </button>
    </div>
    <pre class="iptrc-pre">{{c.output}}</pre>
</div>
<div ng-if="c.error" class="alert alert-danger">{{c.error}}</div>
</div>

<p class="iptrc-disclaimer">
    All outputs are AI-generated drafts. Never paste credentials into any field.
    SMOP outputs are not official MOPs. Verify all steps before execution.
</p>
</div>

```

### Widget – Client Controller (AngularJS)

```

function($scope, $http) {
    var c = this;
    c.tab='ticket'; c.loading=false; c.output='';
    c.error=''; c.copied=false; c.activeSkill='';

    c.ticket = { ticketId:'', platform:'', severity:'',
        symptoms:'', cliDump:'', actionsTaken:'' };
    c.smop = { hostname:'', platform:'', osVersion:'',
        workType:'', windowStart:'', windowEnd:'',
        timezone:'', riskNotes:'' };
    c.rma = { aotsTicket:'', platform:'', problemDescription:'',
        additionalInfo:'', defectivePartFailure:'',
        defectivePidSn:'', hostname:'', ficNumber:'',
        cli:'', oswfTicket:'', shippingAddress:'',
        onsiteContact:'', vendorCase:'', nocTicket:'',
        sameOrNextDay:'', efaRequired:'',
        showChassisFirmware:'', showChassisHardware:'',
        showVersion:'', showLicense:'', showLicenseKeys:'' };
    c.maint = { emailBody:'' };

    c.platforms = ['Cisco IOS-XR','Cisco IOS-XE','Cisco ASR9000',
        'Cisco NCS5500','Juniper Junos','Juniper MX','Arista EOS',
        'Drivenets','Ciena','Nokia SR-OS','Other'];
    c.rmaplatforms = ['Cisco','Juniper','Arista','Ciena',
        'Drivenets','Nokia','Other'];
    c.severities = ['P1 – Complete Outage','P2 – Partial Outage',
        'P3 – Degraded','P4 – Low','Monitoring'];
}

```

```

c.workTypes = ['RE Toggle (Route Engine Switchover)',
  'Optics Replacement', 'PEM Replacement',
  'FPC / Line Card Replacement', 'Software Upgrade',
  'Hardware RMA', 'Configuration Change', 'Other'];
c.cliFields = [
  {label:'show chassis firmware', key:'showChassisFirmware'},
  {label:'show chassis hardware', key:'showChassisHardware'},
  {label:'show version', key:'showVersion'},
  {label:'show system license', key:'showLicense'},
  {label:'show system license keys', key:'showLicenseKeys'}
];

var labels = {
  'ticket-notes':'Ticket Work Notes',
  'smop-draft': 'SMOP/GMOP Scaffold (DRAFT)',
  'rma-email': 'RMA Email – KGPCo Format',
  'maint-email': 'Maintenance Summary + IPTRC Actions'
};

c.rmaInvalid = function() {
  return !c.rma.aotsTicket || !c.rma.platform ||
    !c.rma.problemDescription || !c.rma.defectivePartFailure ||
    !c.rma.defectivePidSn || !c.rma.hostname || !c.rma.clli ||
    !c.rma.oswfTicket || !c.rma.shippingAddress ||
    !c.rma.onsiteContact || !c.rma.sameOrNextDay ||
    !c.rma.efaRequired;
};

/* Adjust base path to match your Scripted REST API namespace */
var BASE = '/api/x_attcs_iptrc/iptrc_ops_api/';

c.runSkill = function(skill, payload) {
  c.loading=true; c.activeSkill=skill;
  c.output=''; c.error=''; c.skillLabel=labels[skill]||skill;
  $http.post(BASE + skill, payload)
    .then(function(res) {
      c.output = res.data.output || '';
      c.loading = false;
    })
    .catch(function(err) {
      c.error = 'Error: ' + (err.data&&err.data.error
        ? err.data.error : 'Request failed.');
      c.loading = false;
    });
};

c.copyOutput = function() {
  var el = document.createElement('textarea');
  el.value = c.output;
  document.body.appendChild(el);
  el.select();
  document.execCommand('copy');
  document.body.removeChild(el);
  c.copied = true;
  setTimeout(function(){
    $scope.$apply(function(){ c.copied=false; });
  }, 2000);
};
}

```

### Widget – Server Script

```
(function() {
```



```
// No server-side processing needed –
// widget calls Scripted REST API directly via $http.
// Optionally pre-populate dropdowns or user info here.
data.user = gs.getUserDisplayName();
})();
```

### Widget – CSS Styles

```
.iptrc-copilot { font-family: 'Open Sans', Arial, sans-serif; }
.iptrc-header { margin-bottom: 16px; }
.iptrc-header h2 { font-size: 22px; font-weight: 700; margin-bottom: 4px; }
.iptrc-panel { padding: 20px 0; }
.iptrc-mono { font-family: Consolas, 'Courier New', monospace;
  font-size: 12px; }
.iptrc-output { margin-top: 20px; border: 1px solid #ddd;
  border-radius: 4px; overflow: hidden; }
.iptrc-output-header {
  display: flex; align-items: center;
  justify-content: space-between;
  padding: 6px 12px; background: #f5f5f5;
  border-bottom: 1px solid #ddd; }
.iptrc-skill-label { font-size: 11px; text-transform: uppercase;
  letter-spacing: .06em; color: #666; font-weight: 600; }
.iptrc-pre { white-space: pre-wrap;
  font-family: Consolas, 'Courier New', monospace;
  font-size: 12px; line-height: 1.6; padding: 16px; margin: 0;
  max-height: 480px; overflow-y: auto; background: #fff; }
.iptrc-cli-box { background: #f9f9f9; border: 1px solid #e8e8e8;
  border-radius: 4px; padding: 12px; margin: 12px 0; }
.iptrc-disclaimer { margin-top: 16px; font-size: 11px; color: #888;
  border-top: 1px solid #eee; padding-top: 8px; }
.spin { animation: spin 1s linear infinite; }
@keyframes spin { 100% { transform: rotate(360deg); } }
```

## System Properties Configuration

Create these 4 System Properties in ServiceNow (sys\_properties.list). Never store the API key as plaintext — use the Credential Store and reference the record ID.

| Property Name       | Value / Example                                                           | Description                          |
|---------------------|---------------------------------------------------------------------------|--------------------------------------|
| iptrc.ai.endpoint   | https://YOUR-RESOURCE.openai.azure.com/openai/deployments/YOUR-DEPLOYMENT | Azure OpenAI deployment endpoint URL |
| iptrc.ai.apikey     | [Credential Store Reference — never plaintext]                            | Azure OpenAI or Anthropic API key    |
| iptrc.ai.model      | gpt-4o                                                                    | Model / deployment name              |
| iptrc.ai.max_tokens | 1500                                                                      | Maximum output tokens per skill call |

■ **IMPORTANT:** For iptrc.ai.apikey, create a Basic Auth or API Key record under Connections & Credentials → Credentials, then reference the credential sys\_id in the system property instead of pasting the raw key.

## Acceptance Criteria (Done Definition)

### Ticket Notes Skill

- Produces consistent headings 100% of time
- Never claims actions not present in user input
- Copy/paste-ready output under 60 seconds end-to-end

### SMOP/GMOP Draft

- Always labeled DRAFT — never claims to be official MOP
- Includes rollback placeholders and evidence capture sections
- Platform-specific command bundles in pre/post checks

### RMA Email Skill

- Always text-only — no tables, no HTML
- Always includes all required KGPCo fields
- Missing fields shown as TBD — never fabricated

### Maintenance Summary

- Always includes timezone quoted exactly from email
- IPTRC Actions checklist produced for every run

- No fabricated confirmation of actions

## Runbook (Operational Ownership)

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- **On-call:** Tools team rotation — same as existing tooling incidents.
- **Bot down:** Fall back to static templates in Teams Files tab. Check ServiceNow system logs → Application Logs → filter by source 'IPTRC'.
- **LLM endpoint down:** Widget will surface error message. No auto-retry — engineer retries manually. Check Azure OpenAI service health.
- **Wrong output / hallucination:** Engineer corrects output. Log the prompt and output as a bug in the tools backlog for prompt engineering review.
- **Secret detected in output:** Immediately delete the output. Review input for credential exposure. Verify `_sanitize()` patterns cover the new pattern.
- **Quality metrics to track:** Regen rate (quality proxy), template compliance pass rate, engineer time-saved self-report (quick thumbs button).

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*IPTRC Ops Copilot v1.0 — AT&T; Internal — Generated 2026-05-28 — Owner: NKAWULA, LEKPAM M — DRAFT — Review before deployment*